

Cats vs Dogs — Final Report (Text-only)

Date: 2025-12-25

Dataset Stats

Item	Value
Dataset	Kaggle Cats vs Dogs (PetImages)
Classes	Cat, Dog
Cat images	12,490
Dog images	12,469
Total images	24,959
Train/Val split	80% / 20% (directory split w/ seed=42)
Image size	224 × 224
Batch size	32
Validation batches used	156 (≈ 4,991 images)

Model Summary

- Backbone: EfficientNetB0 (ImageNet pre-trained), frozen for baseline; fine-tuned last 20 layers.
- Head: GlobalAveragePooling2D → Dropout(0.2) → Dense(1, sigmoid).
- Total params: 4,050,852 | Trainable: 1,281 | Non-trainable: 4,049,571

Training Results

- Training strategy: Transfer learning baseline (Adam lr=1e-3) with early stopping on val AUC; then fine-tuning (Adam lr=1e-4) for last 20 backbone layers.
- Best checkpoint (by val AUC): epoch 2 of baseline training.
- Baseline (best epoch): val_accuracy ≈ 0.9924, val_auc ≈ 0.99939, val_loss ≈ 0.0251
- Fine-tune (best epoch): val_accuracy ≈ 0.9912, val_auc ≈ 0.9994, val_loss ≈ 0.0225

Evaluation — Threshold = 0.50

ROC AUC: 0.999529

Confusion Matrix (rows=true [Cat, Dog], cols=pred [Cat, Dog])

```
[[2428  29]
 [  15 2519]]
```

Classification Report

	precision	recall	f1-score	support
Cat	0.99	0.99	0.99	2457
Dog	0.99	0.99	0.99	2534
accuracy			0.99	4991

macro avg	0.99	0.99	0.99	4991
weighted avg	0.99	0.99	0.99	4991

Threshold Tuning

- Threshold sweep: $t \in [0.05, 0.95]$ step 0.01 on $\text{prob}(\text{dog})$.
- Best threshold (F1 and Accuracy): $t = 0.84$
- At $t=0.84$: Accuracy ≈ 0.99319 , F1 ≈ 0.99326
- Error trade-off: fewer Cat→Dog false positives (7) but more Dog→Cat false negatives (27).

Evaluation — Best F1 Threshold = 0.84

Confusion Matrix (rows=true [Cat, Dog], cols=pred [Cat, Dog])

```
[[2450    7]
 [  27 2507]]
```

Classification Report

	precision	recall	f1-score	support
Cat	0.99	1.00	0.99	2457
Dog	1.00	0.99	0.99	2534
accuracy			0.99	4991
macro avg	0.99	0.99	0.99	4991
weighted avg	0.99	0.99	0.99	4991

Saved Model Paths

- `/kaggle/working/cats_vs_dogs_best.keras` (best by `val_auc`)
- `/kaggle/working/cats_vs_dogs_final.keras` (final saved model)